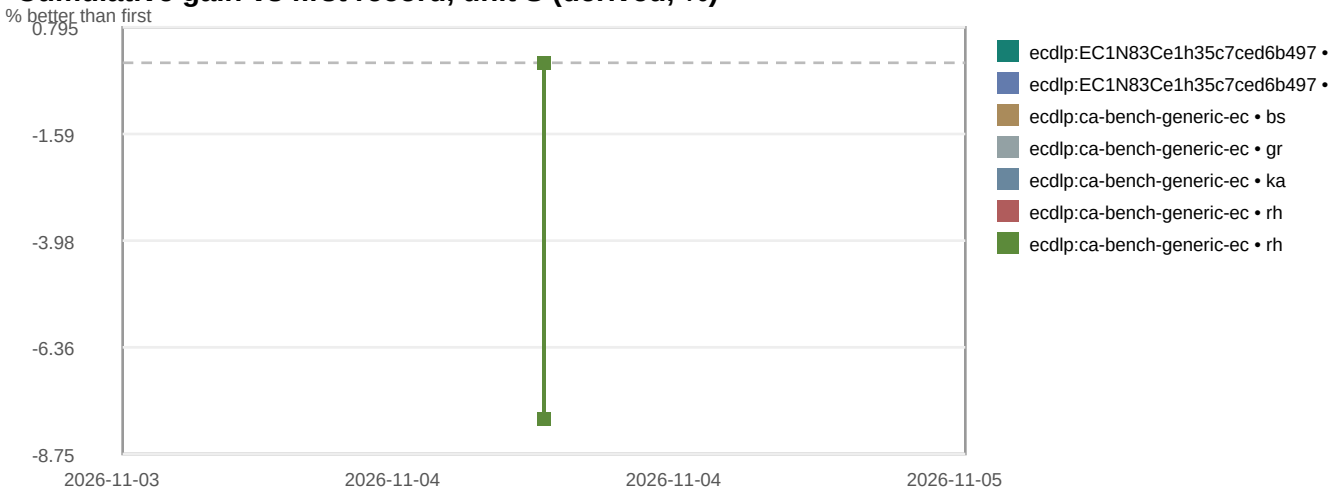
Benchmark progress: cryptanalysis

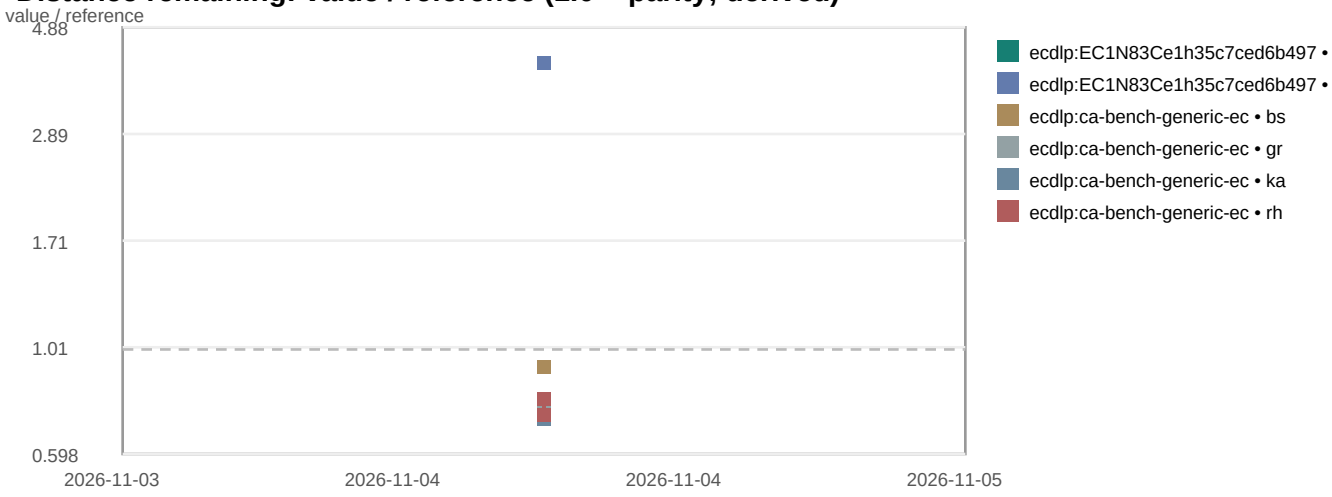
From bench.record/v1 records in Git; raw data in S3 by SHA-256. Derived numbers are labelled derived.

Records: open 0, measured 9, invalid 0, censored 0. Headline unit S: group operations / sqrt(n); the ECDLP cost unit, every phase priced.

Cumulative gain vs first record, unit S (derived, %)

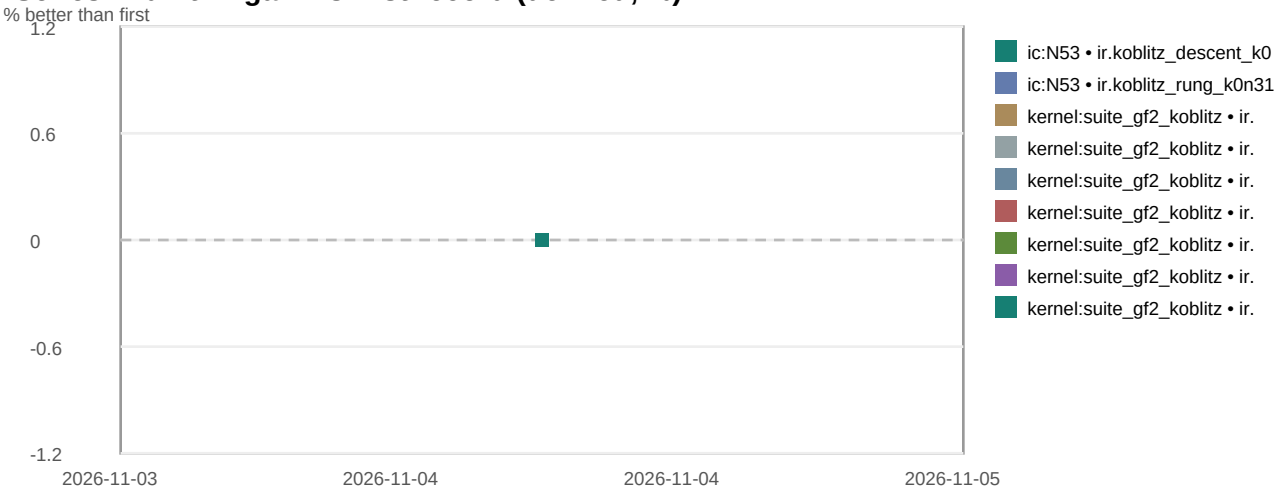


Distance remaining: value / reference (1.0 = parity; derived)

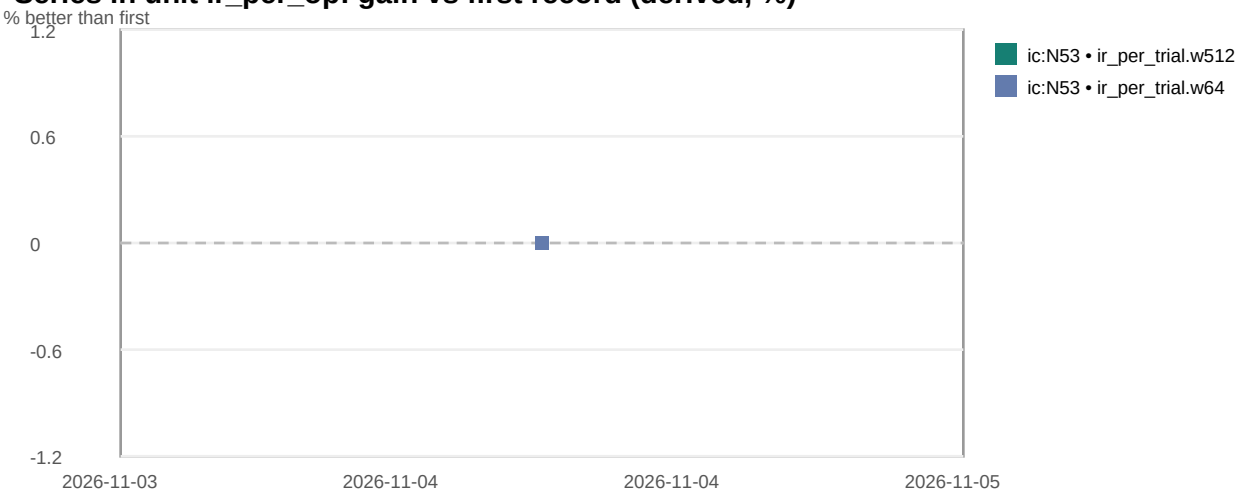


Series in unit count: gain vs first record (derived, %)

Series in unit ir: gain vs first record (derived, %)

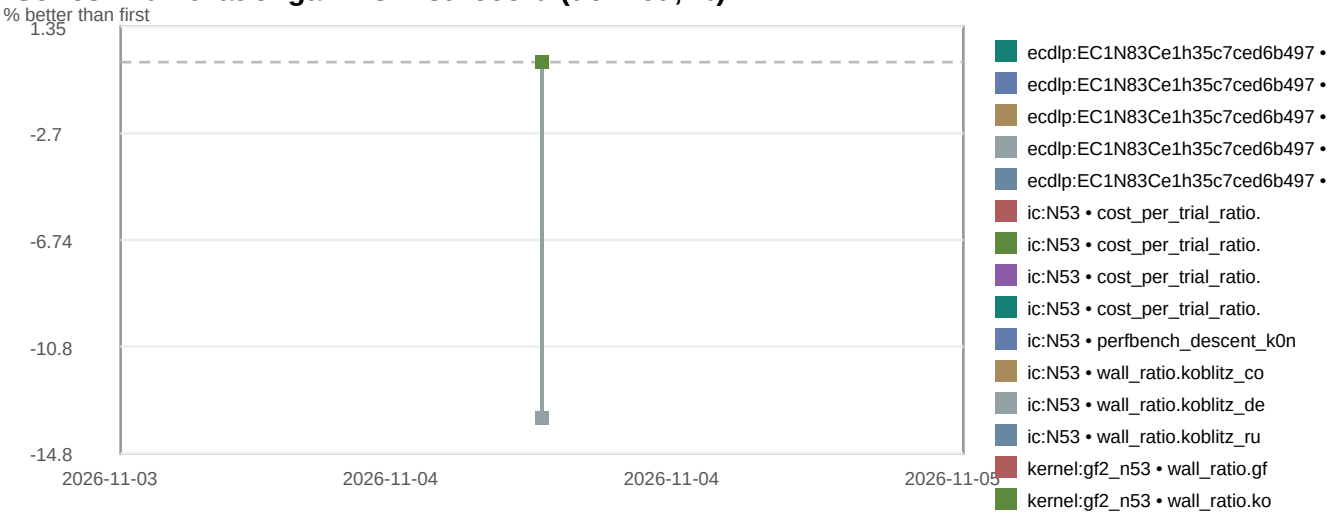


Series in unit ir_per_op: gain vs first record (derived, %)

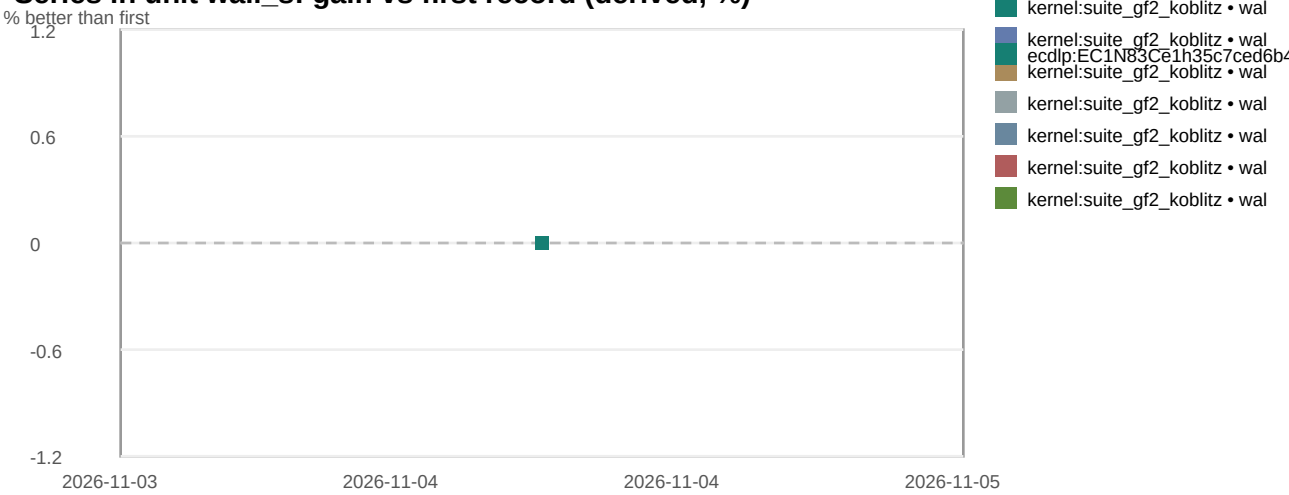


Series in unit percent: gain vs first record (derived, %)

Series in unit ratio: gain vs first record (derived, %)



Series in unit wall_s: gain vs first record (derived, %)



Series summary

problem • metric	unit	first	latest	gain vs first	value/ref	pts
ecdlp:EC1N83Ce1h35c7ced6b497 • e2e_cold_wall_b	wall_s	8.976	8.976	0%	4.0949	1
ecdlp:EC1N83Ce1h35c7ced6b497 • e2e_paired_a_over	ratio	1.261	1.261	-0%	?	1
ecdlp:EC1N83Ce1h35c7ced6b497 • s_e2e_cold	S	0.3922	0.3922	0%	4.0955	1
ecdlp:EC1N83Ce1h35c7ced6b497 • s_online	S	0.0075	0.0075	0%	?	1
ecdlp:EC1N83Ce1h35c7ced6b497 • stage_index_build	ratio	1.3069	1.3069	-0%	?	1
ecdlp:EC1N83Ce1h35c7ced6b497 • stage_online_a_ov	ratio	1.4053	1.4053	-0%	?	1
ecdlp:EC1N83Ce1h35c7ced6b497 • stage_rank_stage_	ratio	1.1996	1.1996	-0%	?	1
ecdlp:EC1N83Ce1h35c7ced6b497 • stage_scan_selfte	ratio	1.3165	1.3165	-0%	?	1
ecdlp:ca-bench-generic-ec • bsgs_s_ec40	S	1.5729	1.5729	0%	0.9183	2
ecdlp:ca-bench-generic-ec • grumpy_s_ec40	S	1.2514	1.2514	0%	0.7306	2
ecdlp:ca-bench-generic-ec • kangaroo_s_ec40	S	1.2201	1.2201	0%	0.7123	2
ecdlp:ca-bench-generic-ec • rho1_s_ec40	S	1.7129	1.7129	0%	?	2
ecdlp:ca-bench-generic-ec • rhot_s_ec40	S	1.2415	1.3402	-7.9508%	0.7824	2
ic:N53 • cost_per_trial_ratio.aa	ratio	0.9918	0.9918	-0%	?	1
ic:N53 • cost_per_trial_ratio.w128	ratio	1.0391	1.0391	-0%	?	1
ic:N53 • cost_per_trial_ratio.w256	ratio	1.0834	1.0834	-0%	?	1
ic:N53 • cost_per_trial_ratio.w512	ratio	1.0946	1.0946	-0%	?	1
ic:N53 • descents_verified	count	3,840	3,840	?	?	1
ic:N53 • filter_admit_rate	percent	12.29	12.29	0%	?	1
ic:N53 • ir.koblitz_descent_k0n53_m2_x4	ir	5.73e+08	5.73e+08	0%	?	1
ic:N53 • ir.koblitz_rung_k0n31	ir	2.026e+08	2.026e+08	0%	?	1
ic:N53 • ir_per_trial.w512	ir_per_op	692.5	692.5	0%	?	1
ic:N53 • ir_per_trial.w64	ir_per_op	735.3	735.3	0%	?	1
ic:N53 • perfbench_descent_k0n53_ratio	ratio	1.2224	1.2224	-0%	?	1
ic:N53 • share.add_many_lazy	percent	38.5	38.5	0%	?	1
ic:N53 • share.contains_key_loop	percent	35.3	35.3	0%	?	1
ic:N53 • share.keys_of	percent	10.6	10.6	0%	?	1
ic:N53 • trials	count	1.207e+07	1.207e+07	?	?	1
ic:N53 • wall_ratio.koblitz_collect_k0n53_w477_u	ratio	1.136	1.136	-0%	?	1
ic:N53 • wall_ratio.koblitz_descent_k0n53_m2_x4	ratio	1.254	1.085	-13.4769%	?	2
ic:N53 • wall_ratio.koblitz_rung_k0n31	ratio	1.151	1.151	-0%	?	1
kernel:gf2_n53 • wall_ratio.gf2_n53_mul_sqr_1m	ratio	0.794	0.794	-0%	?	1
kernel:gf2_n53 • wall_ratio.koblitz_collect_k0n5	ratio	0.865	0.865	-0%	?	1
kernel:suite_gf2_koblitz • ir.gf2_n53_inv_16k	ir	1.978e+07	1.978e+07	0%	?	1
kernel:suite_gf2_koblitz • ir.koblitz_collect_k0	ir	3.147e+08	3.147e+08	0%	?	1
kernel:suite_gf2_koblitz • ir.koblitz_collect_k0	ir	3.261e+08	3.261e+08	0%	?	1
kernel:suite_gf2_koblitz • ir.koblitz_curve_mul_	ir	5.401e+07	5.401e+07	0%	?	1
kernel:suite_gf2_koblitz • ir.koblitz_descent_k0	ir	7.312e+08	7.312e+08	0%	?	1
kernel:suite_gf2_koblitz • ir.koblitz_rung_k0n31	ir	2.314e+08	2.314e+08	0%	?	1
kernel:suite_gf2_koblitz • ir.koblitz_signed_rho	ir	3.364e+08	3.364e+08	0%	?	1
kernel:suite_gf2_koblitz • perfindex_wall_index	ratio	1.2148	1.2148	-0%	?	1
kernel:suite_gf2_koblitz • wall_ratio.gf2_n53_in	ratio	4.1436	4.1436	-0%	?	1
kernel:suite_gf2_koblitz • wall_ratio.koblitz_co	ratio	1.1933	1.1933	-0%	?	1
kernel:suite_gf2_koblitz • wall_ratio.koblitz_co	ratio	1.0116	1.0116	-0%	?	1
kernel:suite_gf2_koblitz • wall_ratio.koblitz_cu	ratio	1.4638	1.4638	-0%	?	1
kernel:suite_gf2_koblitz • wall_ratio.koblitz_de	ratio	1.2539	1.2539	-0%	?	1
kernel:suite_gf2_koblitz • wall_ratio.koblitz_ru	ratio	1.1078	1.1078	-0%	?	1
kernel:suite_gf2_koblitz • wall_ratio.koblitz_si	ratio	1.1986	1.1986	-0%	?	1

Every value is quoted from the record it names, in that record's unit. Percentages and ratios under 'derived' are computed here from those values and are not measurements. Series in different units are never compared.